Power-factor correction, 3-phase, equivalent-circuit tests

Answers and worked solutions · Day 02 · Sun 25 Oct · pack M2

5. Practice

From your course files

Jun23 Q1a (past exam, 5 marks).

> *Hint:* $R_S = (13.8 - 9.5)/300$.
>
> i. On open circuit $I = 0$, so there is no drop across R_S: $E = V_{oc} =$ **13.8 V**.
>
> ii. $R_S = (E - V)/I = (13.8 - 9.5)/300 =$ **0.01433 Ω**. Loss: $I^2 R_S = 300^2 \times 0.01433 =$ **1290 W**. To the motor: $VI = 9.5 \times 300 =$ **2850 W**. Check: $EI = 4140$ W $= 1290 + 2850$.
>
> iii. $V = E - IR_S = 13.8 - 20 \times 0.01433 =$ **13.51 V**.
>
> These match the official answers.

Jun23 Q1b (past exam, 6 marks).

> *Hint:* delta power = 3 × star power.
>
> i. Star: $V_1 = 460/\sqrt{3} = 265.6$ V, so $I_1 = 265.6/70 =$ **3.794 A**, $I_L = I_1 =$ **3.794 A**, and $P = 3V_1^2/R = 3 \times 265.6^2/70 =$ **3.023 kW**.
>
> ii. Delta: $V_1 = V_L = 460$ V, so $I_1 = 460/70 =$ **6.571 A**, $I_L = \sqrt{3} I_1 =$ **11.38 A**, and $P = 3 \times 460^2/70 =$ **9.069 kW** (exactly 3 times the star power).
>
> These match the official answers.

Jun24 Q1d (past exam, 8 marks).

> *Hint:* (iii) uses $Q_C = -30$ VAr, 115 V, **60 Hz**.
>
> i. Triangle: $P = 80$ W horizontal, $Q = +60$ VAr up, so $S = 100$ VA. pf $= 80/100 =$ **0.8 lag**.
>
> ii. The capacitor takes no real power, so $P_S =$ **80 W**.
> $Q_S = P_S \tan(\cos^{-1} 0.9) = 80 \times 0.4843 =$ **38.75 VAr**.
> $Q_C = Q_S - Q_L = 38.75 - 60 =$ **−21.25 VAr**.

iii. $C = \dfrac{|Q_C|}{\omega V^2} = \dfrac{30}{2\pi(60) \times 115^2}$ = **6.017 µF**. (Using −21.25 VAr from part (ii) gives 4.26 µF, which is wrong.)

These match the official answers.

Power Circuit Tool WS2 Task 1 (workshop).

Hint: assume the tool's 240 V, 50 Hz supply. The workshop solutions mention a 240 V source; 50 Hz is assumed. If your simulator showed a different frequency, rescale C by $50/f$. (The official sheet gives only qualitative answers.)

$I = (S/V)^* = \big((P + jQ)/240\big)^*.$

Case	P	Q	I_{supply}
1	100	0	$0.4167\angle 0^\circ$ A
2	0	100	$0.4167\angle -90^\circ$ A
3	100	100	$0.5893\angle -45^\circ$ A

1. Same magnitude, but case 2 lags by 90°.
2. Same fluctuating part at 2f, but case 2 has zero average power.
3. $I_3 = I_1 + I_2$ (superposition: P and Q add).

Power Circuit Tool WS2 Task 2 (workshop).

$P = 100$ W and $Q = 200$ VAr, so $S = 223.6$ VA and $I = 0.9317\angle -63.43^\circ$ A (the same for both models, because the tool fixes P and Q).

Case	I_{supply}	R (Ω)	X (Ω)
Series	$0.9317\angle -63.43^\circ$ A	$P/I^2 = 115.2$	$Q/I^2 = 230.4$
Parallel	$0.9317\angle -63.43^\circ$ A	$V^2/P = 576$	$V^2/Q = 288$

1. Series: the **current** is common.
2. R:X = P:Q = **1:2**, because $P = I^2R$ and $Q = I^2X$ share the same I^2.
3. Parallel: the **voltage** is common.
4. R:X = Q:P = **2:1**, because $R = V^2/P$ and $X = V^2/Q$ share the same V^2.

Power Circuit Tool WS2 Task 3 (workshop).

$P = 100$ W, $Q = 140$ VAr. The exact optimum is $C = 140/(314.16 \times 240^2) = 7.74$ µF. With integer steps, 7 µF leaves +13.3 VAr and 8 µF leaves −4.8 VAr, so C_{opt} = **8 µF**.

Case	C (µF)	P_S (W)	Q_S (VAr)	I_S (A)	pf
1	0	100	+140.0	$0.7169\angle -54.46^\circ$	0.581 lag
2	4	100	+67.6	$0.5030\angle -34.07^\circ$	0.828 lag
3	8	100	−4.8	$0.4171\angle +2.73^\circ$	0.999 lead
4	12	100	−77.1	$0.5262\angle +37.65^\circ$	0.792 lead

1. P_S stays at 100 W because a capacitor absorbs no real power.
2. From 0 to C_{opt} the magnitude of I_S falls and its angle rises towards 0. Beyond C_{opt} the magnitude rises again and the angle becomes positive (leading).
3. At C_{opt}, $I_S = 0.4171$ A. The minimum possible current for 100 W at unity pf is $100/240 = 0.4167$ A, almost identical.

PQ 2.20 (practice questions).

Hint: 60 Hz. This is also worked in your Test Reference 6.1.

a. $I_1 = 200/100 =$ **2 A**. $|Z_2| = 22.36\ \Omega$, so $I_2 = 200/22.36 =$ **8.944 A**.
$P = 2^2(100) + 8.944^2(10) = 400 + 800 =$ **1200 W** and $Q = 8.944^2(20) =$ **+1600 VAr**. $S = 2000$ VA, so pf = **0.600 lag**.

b. For pf 0.8 lag at the same P: $Q_{\text{new}} = 1200\tan(\cos^{-1}0.8) = +900$ VAr, so $Q_C = 900 - 1600 = -700$ VAr. $X_C = V^2/Q_C = 200^2/(-700) = -57.14\ \Omega$, so $C = 1/(2\pi \cdot 60 \cdot 57.14) =$ **46.42 µF**.

c. Before: **2000 VA**. After: $\sqrt{1200^2 + 900^2} =$ **1500 VA**.

d. Both triangles share P = 1200 W. Before: Q = +1600 VAr, S = 2000 VA. After: Q = +900 VAr, S = 1500 VA. The capacitor's −700 VAr is the vertical drop between them.

These match the official answers.

PQ 2.6 (practice questions).

Hint: (d) $I = (E - V)/R$.

a. $V_{oc} = E =$ **12.5 V**.

b. $I_{sc} = 12.5/0.02 =$ **625 A**, $P = I^2R =$ **7.813 kW**.

c. $I = 12.5/1.02 = 12.25$ A, $V = 12.25$ V. The official sheet gives **12.26 V, 12.26 A**; the exact value is 12.2549, so the difference is rounding.

d. $I = (12.5 - 8)/0.02 =$ **225 A**, $P = 8 \times 225 =$ **1.800 kW**.

e. At $R_L = R = 0.02\ \Omega$: $P_{max} = E^2/(4R) =$ **1.953 kW** ($V = 6.25$ V, $I = 312.5$ A).

PQ 2.7 (practice questions).

a. $E = V_{oc} =$ **1.61 V** and $R = V_{oc}/I_{sc} = 1.61/3.1 =$ **0.5194 Ω**.

b. $V = E - IR = 1.61 - 0.5 \times 0.5194 =$ **1.350 V** and $P = VI =$ **0.6752 W**.

c. At $R_L = R$: $P_{max} = E^2/(4R) = 1.61^2/(4 \times 0.5194) =$ **1.248 W**.

These match the official answers.

PQ 2.21 (practice questions).

Hint: series uses I^2; parallel uses V^2.

a. $R = P/I^2 = 25/400 =$ **0.0625 Ω**. $|Z| = 5/20 = 0.25\ \Omega$, so $X = \sqrt{0.25^2 - 0.0625^2} =$ **0.2421 Ω**.

b. $R = V^2/P = 110^2/40 =$ **302.5 Ω**. $S = 110 \times 2.4 = 264$ VA, $Q = \sqrt{264^2 - 40^2} = 261.0$ VAr, so $X = V^2/Q =$ **46.36 Ω** (46.37 Ω unrounded).

These match the official answers.

PQ 2.26 (practice questions).

Hint: draw the star and delta first.

a. Star: $V_L = \sqrt{3} \times 110 =$ **190.5 V**, $I_L = I_1 =$ **10 A**.

b. Delta: $V_L = V_1 =$ **110 V**, $I_L = \sqrt{3} \times 10 =$ **17.32 A**.

These match the official answers.

PQ 2.28 (practice questions).

a. $V_1 = 230/\sqrt{3} = 132.8$ V, so $R = 132.8/10 =$ **13.28 Ω**.

b. Delta: $V_1 =$ **230 V**, $I_1 = 230/13.28 =$ **17.32 A**, $I_L = \sqrt{3} \times 17.32 =$ **30 A**.

c. Phase voltage ×$\sqrt{3}$, phase current ×$\sqrt{3}$, supply (line) current ×**3**.

These match the official answers. (The text copy of the answer sheet loses the √ symbols; the PDF itself shows √3, √3 and 3.)

PQ 2.30 (practice questions).

Hint: your Test Reference 7.2.

a. Delta: $V_1 = V_L =$ **415 V**.

b. $Z = 2 + j3 = 3.606\angle 56.31^\circ\ \Omega$, so $I_1 = 415/3.606\angle 56.31^\circ =$ **115.1∠−56.31° A**.

c. $I_L = \sqrt{3} \times 115.1 =$ **199.4 A**.

d. pf $= \cos 56.31^\circ =$ **0.5547 lag**.

e. $S = 3V_1 I_1 = 3 \times 415 \times 115.1 = 143.3$ kVA. The official answer is **143.2 kVA**; the difference is rounding.

f. $P = S \times \text{pf} =$ **79.50 kW** (79.49 kW unrounded).

g. $3I_1^2 R = 3 \times 115.1^2 \times 2 =$ **79.50 kW**, the same.

h. $Q = 3I_1^2 X = 3 \times 115.1^2 \times 3 =$ **+119.2 kVAr**.

PQ 2.31 (practice questions).

In star each phase sees $V_1 = 415/\sqrt{3} = 239.6$ V instead of 415 V.

a. phase voltage ×$1/\sqrt{3}$ (239.6 V). (b) phase current ×$1/\sqrt{3}$ (66.45 A, same angle). (c) line current ×1/3: in star $I_L = I_1 = 66.45$ A, versus 199.4 A before. (d) pf ×1 (the impedance angle is unchanged). (e)–(h) S, P (both ways) and Q all ×1/3, because each goes as V_1^2 per phase.

Check: $P = 3 \times 66.45^2 \times 2 = 26.50$ kW $= 79.50/3$.

These match the official answers (the PDF shows $1/\sqrt{3}$ for (a) and (b); the text copy loses the √).

Jun23 Q4b (past exam, 7 marks), stretch.

Hint: the synchronous motor *generates* its spare vars: $Q_{\text{syn}} = -\sqrt{250^2 - 222.2^2}$.

i. $P_{im} = 200/0.85 =$ **235.3 kW** and $Q_{im} = 235.3 \tan(\cos^{-1} 0.83) =$ **+158.1 kVAr**.

ii. $P_{syn} = 200/0.90 =$ **222.2 kW**. Its 250 kVA rating limits S, so the most reactive power it can supply is $\sqrt{250^2 - 222.2^2} =$ **114.6 kVAr** (generated, so $Q_{syn} = -114.6$ kVAr). Unrounded this is 114.5 kVAr.

iii. $P = 235.3 + 222.2 = 457.5$ kW and $Q = 158.1 - 114.6 = 43.5$ kVAr, so the best plant pf is $457.5/\sqrt{457.5^2 + 43.5^2} =$ **0.9955 lag**.

These match the official answers.

New questions

N1.

a. Delta: $V_1 = V_L = 400$ V, so $I_1 = 400/48 =$ **8.333 A**, $I_L = \sqrt{3} \times 8.333 =$ **14.43 A**, and $P = 3V_1^2/R = 3(400^2)/48 =$ **10.0 kW**.

b. Star: $V_1 = 400/\sqrt{3} =$ **230.9 V**, $I_L = I_1 = 230.9/48 =$ **4.811 A**, and $P = 3(230.9^2)/48 =$ **3.333 kW**. The ratio is $P_\Delta/P_Y =$ **3**.

N2.

a. The drop rises by $12.6 - 11.2 = 1.4$ V for an extra 70 A, so $R_S = 1.4/70 =$ **0.02 Ω**. Then $E = 12.6 + 5(0.02) =$ **12.7 V**.

b. $V = 12.7 - 150(0.02) =$ **9.7 V**, $P_{\text{load}} = 9.7 \times 150 =$ **1455 W**, and $P_{\text{loss}} = 150^2 \times 0.02 =$ **450 W**. Check: $EI = 1905$ W $= 1455 + 450$.

c. $P_{max} = E^2/(4R_S) = 12.7^2/0.08 =$ **2016 W**, at $I = 317.5$ A and $V = 6.35$ V.

N3.

a. $S = 230 \times 6.2 =$ **1426 VA**, $Q = +\sqrt{1426^2 - 1100^2} =$ **+907.5 VAr** (positive because a motor is inductive), and pf $= 1100/1426 =$ **0.7714 lag**.

b. $R_S = P/I^2 = 1100/6.2^2 =$ **28.62 Ω**, $X_S = Q/I^2 = 23.61\ \Omega$, so $L_S = 23.61/314.16 =$ **75.1 mH**.

c. $R_P = V^2/P = 230^2/1100 =$ **48.09 Ω**, $X_P = V^2/Q = 58.29\ \Omega$, so $L_P =$ **185.6 mH**.

d. $C = 907.5/(314.16 \times 230^2) =$ **54.6 µF**, and $I = P/V = 1100/230 =$ **4.78 A** (down from 6.2 A).

N4.

a. Motor input: $P_{im} = 22/0.9 = 24.44$ kW and $Q_{im} = 24.44 \tan(\arccos 0.82) = +17.06$ kVAr. The heater adds 10 kW and 0 kVAr.

$$P = 34.44\ \text{kW}, \quad Q = +17.06\ \text{kVAr}, \quad S = 38.44\ \text{kVA}, \quad \textbf{pf} = \textbf{0.8961 lag}, \quad I_L = \frac{38440}{\sqrt{3} \cdot 415} = \textbf{53.48 A}$$

b. $Q_S = 34.44 \tan(\arccos 0.98) = 6.994$ kVAr, so $Q_C = 6.994 - 17.06 =$ **−10.07 kVAr** (−3.356 kVAr per capacitor).

c. Delta: $C_\Delta = 3356/(314.16 \times 415^2) =$ **62.0 µF**, rated 415 V. Star: $C_Y = 3356/(314.16 \times 239.6^2) =$ **186.1 µF**, rated 240 V. The delta bank needs the higher voltage rating. The new line current is $35.15\ \text{kVA}/(\sqrt{3} \cdot 415) = 48.90$ A.

d. In star each 62.0 µF capacitor sees only 239.6 V, so the bank supplies $Q_C/3 = -3.356$ kVAr. Then $Q_S = 17.06 - 3.356 = 13.71$ kVAr and pf $= 34.44/\sqrt{34.44^2 + 13.71^2} =$ **0.929 lag**.

N5.

a. $S = \sqrt{25^2 + 31^2} = 39.82$ VA, $I = 39.82/60.5 =$ **0.658 A** (Table 5 measured $I_L \approx$ 0.658 A), and pf $= 25/39.82 =$ **0.628 lag**.

b. $\tan(\arccos 0.9) = 0.4843$, so $|Q_S| = 12.11$ VAr.

- Lagging: $Q_S = +12.11$ gives $Q_C = -18.89$ VAr and $C = 18.89/(314.16 \times 60.5^2) =$ **16.4 µF (lag)**.
- Leading: $Q_S = -12.11$ gives $Q_C = -43.11$ VAr and $C =$ **37.5 µF (lead)**.

c. Both give $S = 25/0.9 = 27.78$ VA, so $I_S =$ **0.459 A** in both cases. The minimum current is at unity pf: $25/60.5 =$ **0.413 A**.

d. Reactive power is conserved: $Q_S = Q_L + Q_C = 31.0 - 45.3 = -14.3$ VAr. That is negative, so the supply value must be **−16.6 VAr**. The circuit is over-corrected and the 0.837 pf is **leading**.

6. Quick recall check

Question

1. $415/\sqrt{3} = 239.6$ V.
2. Both rise by a factor of 3. The pf does not change.
3. $S = 100$ VA and $Q = +80$ VAr, so $Q_C = -80$ VAr.
4. $(12.0 - 11.4)/30 = 0.02\ \Omega$.

5. $C_Y = 3C_\Delta$. The star bank needs three times the capacitance, at $1/\sqrt{3}$ of the voltage rating.